

PostgreSQL Complex Case Study

Multi-Warehouse E-Commerce and Supply Chain Management System

1. Business Background

NexaCommerce Pvt. Ltd. is a large e-commerce company selling electronics, fashion products, home appliances, furniture, books, and office equipment across India.

The company operates:

- Multiple regional warehouses
- Multiple product suppliers
- Online customer accounts
- Product catalogues with categories and brands
- Warehouse-level inventory
- Customer orders containing multiple products
- Discount campaigns and promotional coupons
- Multiple payment attempts per order
- Partial and complete shipments
- Product returns and refunds
- Delivery partners
- Employee and manager hierarchies
- Supplier purchase orders
- Stock transfers between warehouses

Currently, warehouse stock, customer orders, supplier information, payment details, and product returns are maintained in separate systems.

This has created several operational problems:

- Customers place orders for products that are not actually available.
- Warehouse stock is not updated consistently.
- A single order cannot be tracked across multiple shipments.
- Payment failures and repeated payment attempts are difficult to monitor.
- Refunds are not properly linked to returned products.
- Supplier deliveries are not compared with ordered quantities.
- Stock transferred between warehouses is difficult to trace.
- Management cannot identify profitable products, warehouses, suppliers, or customers.
- Duplicate customer and supplier records are being created.
- Reports are prepared manually using spreadsheets.

NexaCommerce wants to implement a centralized PostgreSQL-based **E-Commerce and Supply Chain Management System**.

2. Main Objective

Design and implement a normalized PostgreSQL database that supports the complete business flow from product procurement to customer delivery and return processing.

The database must support:

1. Product catalogue management
2. Supplier management
3. Warehouse management
4. Warehouse-level inventory
5. Customer account management
6. Shopping and order processing
7. Order-item level pricing and discounts
8. Payment processing
9. Shipment tracking
10. Product returns
11. Refund processing
12. Purchase order management
13. Supplier delivery tracking
14. Warehouse stock transfers
15. Employee-manager hierarchy
16. Operational and management reporting
17. Data import and export
18. Backup and restoration

3. User Roles

The system will be used by the following roles.

3.1 Customer

A customer can:

- Register an account
- Maintain contact and delivery addresses
- Browse products
- Place orders
- Use promotional coupons
- Make payments
- Track shipments
- Request returns

- View refund status
- Review purchased products

3.2 Sales Executive

A sales executive can:

- View customer orders
- Verify customer information
- Update order status
- Coordinate with warehouse teams
- Handle order-related customer requests

3.3 Warehouse Executive

A warehouse executive can:

- Check product stock
- Reserve inventory
- Pack order items
- Create shipments
- Confirm dispatched quantities
- Receive supplier deliveries
- Process stock transfers
- Accept returned products

3.4 Warehouse Manager

A warehouse manager can:

- Approve stock adjustments
- Approve stock transfers
- Monitor low-stock products
- Verify damaged stock
- Review warehouse performance
- Manage warehouse employees

3.5 Procurement Executive

A procurement executive can:

- Create supplier purchase orders
- Add products to purchase orders
- Track expected delivery dates
- Verify received quantities
- Monitor delayed suppliers

3.6 Finance Executive

A finance executive can:

- View payment attempts
- Confirm payment settlements
- Process refunds
- Monitor outstanding amounts
- Generate revenue and tax reports

3.7 System Administrator

The administrator can:

- Manage users and employees
 - Maintain master data
 - Import and export records
 - Perform backup and restore operations
 - Monitor database quality
 - Manage inactive records
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4. Required Database Entities

The database should contain at least the following entities.

4.1 Customers

Store information such as:

- Customer ID
- Customer name
- Email
- Phone number
- Registration date
- Customer type
- Loyalty points
- Account status
- Last login timestamp

The email must be unique.

A customer may have multiple addresses.

4.2 Customer Addresses

Store:

- Address ID
- Customer ID
- Address type
- Address line
- City
- State
- Postal code
- Country
- Default address indicator

Address types may include:

- Home
- Office
- Billing
- Shipping

Only one address should be marked as the default shipping address for a customer.

4.3 Product Categories

Store:

- Category ID
- Category name
- Parent category ID
- Description
- Active status

The category structure must support subcategories.

Examples:

- Electronics
- Mobiles
- Laptops
- Accessories
- Home Appliances
- Kitchen Appliances
- Air Conditioners

This table should use a self-referencing relationship.

4.4 Brands

Store:

- Brand ID
- Brand name
- Country of origin
- Support email
- Active status

A brand may contain multiple products.

4.5 Products

Store:

- Product ID
- Product code
- Product name
- Category ID
- Brand ID
- Product description
- Selling price
- Standard cost
- Tax percentage
- Product weight
- Warranty duration
- Reorder level
- Created timestamp
- Active status

The product code must be unique.

Selling price and standard cost cannot be negative.

4.6 Suppliers

Store:

- Supplier ID
- Supplier name
- Contact person
- Email

- Phone
- City
- State
- GST or tax registration number
- Supplier rating
- Active status
- Registration timestamp

Supplier email and tax registration number should be unique.

4.7 Supplier Products

Create a relationship between suppliers and products.

Store:

- Supplier ID
- Product ID
- Supplier product code
- Purchase price
- Minimum order quantity
- Average delivery days
- Last supplied date

A product may be supplied by multiple suppliers.

A supplier may supply multiple products.

4.8 Warehouses

Store:

- Warehouse ID
- Warehouse name
- Warehouse code
- City
- State
- Capacity
- Opening date
- Manager ID
- Active status

The warehouse code must be unique.

4.9 Employees

Store:

- Employee ID
- Employee name
- Email
- Designation
- Salary
- Joining date
- Warehouse ID
- Department
- Manager ID
- Employment status

The employee table must support employee-manager hierarchy using a self-reference.

An employee may work in:

- Sales
- Warehouse
- Procurement
- Finance
- Customer Support
- Administration

4.10 Warehouse Inventory

Store warehouse-level stock details:

- Warehouse ID
- Product ID
- Available quantity
- Reserved quantity
- Damaged quantity
- Reorder level
- Last stock update timestamp

The combination of warehouse and product must be unique.

Available quantity, reserved quantity, and damaged quantity cannot be negative.

4.11 Customers' Orders

Store:

- Order ID
- Customer ID
- Billing address ID
- Shipping address ID
- Order date and time
- Order status
- Gross amount
- Discount amount
- Tax amount
- Delivery charge
- Final amount
- Coupon ID
- Sales executive ID
- Order source

Order sources may include:

- Website
- Mobile Application
- Customer Support
- Marketplace

Order statuses may include:

- Created
- Confirmed
- Processing
- Partially Shipped
- Shipped
- Delivered
- Cancelled
- Partially Returned
- Returned

4.12 Order Items

Each order can contain multiple products.

Store:

- Order item ID
- Order ID

- Product ID
- Warehouse ID
- Quantity
- Unit price
- Discount percentage
- Tax percentage
- Net amount
- Item status

Item statuses may include:

- Ordered
- Reserved
- Packed
- Shipped
- Delivered
- Cancelled
- Returned

The product price recorded in the order item must represent the price at the time of purchase.

4.13 Coupons

Store:

- Coupon ID
- Coupon code
- Description
- Discount type
- Discount value
- Minimum order amount
- Maximum discount amount
- Start date
- End date
- Usage limit
- Customer usage limit
- Active status

Discount types may include:

- Fixed Amount
- Percentage

The coupon code must be unique.

4.14 Coupon Usage

Store:

- Coupon usage ID
- Coupon ID
- Customer ID
- Order ID
- Usage timestamp
- Discount amount

This table should be used to enforce coupon usage limits.

4.15 Payments

A customer may make multiple payment attempts for the same order.

Store:

- Payment ID
- Order ID
- Payment method
- Transaction reference
- Payment amount
- Payment status
- Payment timestamp
- Failure reason
- Payment gateway

Payment methods may include:

- UPI
- Credit Card
- Debit Card
- Net Banking
- Wallet
- Cash on Delivery

Payment statuses may include:

- Initiated
- Pending
- Successful
- Failed
- Cancelled
- Refunded

- Partially Refunded

The transaction reference should be unique when available.

4.16 Delivery Partners

Store:

- Delivery partner ID
- Company name
- Contact number
- Email
- Service type
- Active status

Service types may include:

- Standard
 - Express
 - Same Day
 - Heavy Goods
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4.17 Shipments

An order may be divided into multiple shipments.

Store:

- Shipment ID
- Order ID
- Warehouse ID
- Delivery partner ID
- Tracking number
- Dispatch timestamp
- Expected delivery date
- Actual delivery timestamp
- Shipment status
- Shipping cost

Shipment statuses may include:

- Created
- Packed
- Dispatched
- In Transit

- Out for Delivery
- Delivered
- Delivery Failed
- Returned to Warehouse

The tracking number must be unique.

4.18 Shipment Items

Store:

- Shipment item ID
- Shipment ID
- Order item ID
- Shipped quantity

One order item may be split across multiple shipments.

The total shipped quantity must not exceed the ordered quantity.

4.19 Product Returns

Store:

- Return ID
- Order ID
- Customer ID
- Return request timestamp
- Return reason
- Return status
- Approved by employee ID
- Received warehouse ID
- Received timestamp

Return statuses may include:

- Requested
 - Approved
 - Rejected
 - Pickup Scheduled
 - Received
 - Inspected
 - Refund Initiated
 - Closed
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4.20 Return Items

Store:

- Return item ID
- Return ID
- Order item ID
- Return quantity
- Item condition
- Return reason
- Inspection status
- Restock indicator
- Refund amount

Item conditions may include:

- Unopened
- Good
- Damaged
- Defective
- Incorrect Product

The return quantity must not exceed the delivered quantity.

4.21 Refunds

Store:

- Refund ID
- Return ID
- Payment ID
- Refund amount
- Refund method
- Refund status
- Refund timestamp
- Refund reference

Refund statuses may include:

- Initiated
- Processing
- Successful
- Failed

A refund amount cannot exceed the successfully paid amount after previous refunds.

4.22 Purchase Orders

Store:

- Purchase order ID
- Supplier ID
- Warehouse ID
- Created by employee ID
- Purchase order date
- Expected delivery date
- Purchase order status
- Total amount
- Approved by employee ID
- Approval timestamp

Purchase order statuses may include:

- Draft
 - Submitted
 - Approved
 - Partially Received
 - Received
 - Cancelled
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4.23 Purchase Order Items

Store:

- Purchase order item ID
- Purchase order ID
- Product ID
- Ordered quantity
- Purchase price
- Tax percentage
- Received quantity
- Rejected quantity

Received quantity plus rejected quantity must not exceed ordered quantity.

4.24 Supplier Deliveries

Store:

- Supplier delivery ID

- Purchase order ID
- Delivery reference number
- Delivery date
- Received by employee ID
- Delivery status
- Supplier invoice number
- Supplier invoice amount

A purchase order may have multiple supplier deliveries.

4.25 Supplier Delivery Items

Store:

- Supplier delivery item ID
- Supplier delivery ID
- Purchase order item ID
- Delivered quantity
- Accepted quantity
- Rejected quantity
- Rejection reason

Accepted quantity plus rejected quantity must equal delivered quantity.

4.26 Warehouse Stock Transfers

Store:

- Transfer ID
- Source warehouse ID
- Destination warehouse ID
- Requested by employee ID
- Approved by employee ID
- Request timestamp
- Dispatch timestamp
- Received timestamp
- Transfer status

Transfer statuses may include:

- Requested
- Approved
- Rejected
- Dispatched
- Partially Received

- Received
- Cancelled

The source and destination warehouse cannot be the same.

4.27 Stock Transfer Items

Store:

- Transfer item ID
- Transfer ID
- Product ID
- Requested quantity
- Approved quantity
- Dispatched quantity
- Received quantity
- Damaged quantity

Received quantity plus damaged quantity must not exceed dispatched quantity.

4.28 Product Reviews

Store:

- Review ID
- Customer ID
- Product ID
- Order item ID
- Rating
- Review title
- Review text
- Review timestamp
- Approval status

A customer should only be allowed to review a product purchased and delivered through an order.

Rating must be between 1 and 5.

5. Relationship Requirements

The database design must support the following relationships:

1. One customer can have multiple addresses.

2. One customer can place multiple orders.
 3. One order can contain multiple order items.
 4. One product can appear in multiple order items.
 5. One product belongs to one category and one brand.
 6. One category may have one parent category.
 7. One supplier can supply multiple products.
 8. One product can be supplied by multiple suppliers.
 9. One warehouse can store multiple products.
 10. One product can be available in multiple warehouses.
 11. One order can have multiple payment attempts.
 12. One order can be divided into multiple shipments.
 13. One shipment can contain multiple shipment items.
 14. One order can have multiple return requests.
 15. One return can contain multiple returned items.
 16. One return may generate one or more refund records.
 17. One supplier can receive multiple purchase orders.
 18. One purchase order can contain multiple products.
 19. One purchase order can be received through multiple supplier deliveries.
 20. One warehouse transfer can contain multiple products.
 21. One employee can manage multiple employees.
 22. One warehouse can have multiple employees.
 23. One customer can submit multiple product reviews.
 24. One delivery partner can manage multiple shipments.
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6. Primary Key Requirements

Every table must have a primary key.

Use generated numeric identifiers for transactional tables such as:

- Orders
- Payments
- Shipments
- Returns
- Refunds
- Purchase orders
- Deliveries
- Stock transfers

Composite uniqueness should be applied where required, such as:

- Warehouse and product in warehouse inventory
- Supplier and product in supplier products
- Coupon, customer, and order in coupon usage
- Customer and order item in product reviews

7. Foreign Key Requirements

Foreign keys must ensure referential integrity.

Examples include:

- Orders must reference valid customers.
- Order items must reference valid orders and products.
- Payments must reference valid orders.
- Shipments must reference valid orders, warehouses, and delivery partners.
- Return items must reference valid returns and order items.
- Refunds must reference valid returns and payments.
- Purchase orders must reference valid suppliers and warehouses.
- Stock transfers must reference valid source and destination warehouses.
- Employees may reference another employee as manager.

Appropriate cascading behavior should be defined carefully.

Transaction history should not be deleted accidentally when master records are removed.

8. Data Validation Requirements

The database must enforce the following rules:

1. Product selling price cannot be negative.
2. Product standard cost cannot be negative.
3. Stock quantities cannot be negative.
4. Ordered quantity must be greater than zero.
5. Payment amount must be greater than zero.
6. Refund amount must be greater than zero.
7. Discount percentage must be between 0 and 100.
8. Tax percentage must be between 0 and 100.
9. Rating must be between 1 and 5.
10. Supplier rating must be within an accepted range.
11. Warehouse capacity must be greater than zero.
12. Employee salary must be greater than zero.
13. Coupon start date must be earlier than or equal to the end date.
14. Purchase order expected date cannot be earlier than the purchase order date.
15. A transfer source and destination warehouse cannot be identical.
16. Shipped quantity cannot exceed ordered quantity.
17. Return quantity cannot exceed delivered quantity.
18. Refund amount cannot exceed the eligible refundable amount.
19. Received stock cannot exceed ordered stock.

20. Customer and employee email addresses should follow a valid pattern.
 21. Product code, coupon code, warehouse code, transaction reference, and tracking number should be unique.
 22. Inactive products should not be added to new orders.
 23. Inactive warehouses should not be used for new shipments or transfers.
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9. DDL Requirements

Participants must perform the following structural operations:

1. Create the complete database.
 2. Create all required tables.
 3. Add primary keys.
 4. Add foreign keys.
 5. Add unique constraints.
 6. Add check constraints.
 7. Add default values.
 8. Add timestamp columns.
 9. Create self-referencing relationships.
 10. Add a new column to an existing table.
 11. Change the data type of a column.
 12. Rename a column.
 13. Rename a table.
 14. Add a new constraint after table creation.
 15. Remove a constraint.
 16. Remove a column.
 17. Create a temporary table.
 18. Remove a temporary table.
 19. Create at least two views.
 20. Create indexes for frequently searched columns.
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10. DML Requirements

Participants must perform the following data operations:

1. Insert at least ten categories.
2. Insert at least eight brands.
3. Insert at least fifty products.
4. Insert at least ten suppliers.
5. Map suppliers to products.
6. Insert at least five warehouses.
7. Insert at least twenty employees.
8. Insert at least fifty customers.

9. Insert multiple addresses for selected customers.
 10. Insert warehouse inventory for all products.
 11. Insert at least forty customer orders.
 12. Insert at least one hundred order items.
 13. Insert multiple payment attempts for selected orders.
 14. Insert successful and failed payment records.
 15. Insert partial shipment examples.
 16. Insert delivered and failed shipment records.
 17. Insert product returns.
 18. Insert partial and complete refund records.
 19. Insert supplier purchase orders.
 20. Insert purchase order item details.
 21. Insert partial supplier delivery records.
 22. Insert warehouse stock transfer records.
 23. Insert product reviews.
 24. Update product prices.
 25. Update inventory quantities.
 26. Update order statuses.
 27. Update shipment statuses.
 28. Update employee information.
 29. Delete invalid temporary data.
 30. Delete inactive master records only when no transaction history exists.
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11. Basic Query Requirements

Prepare queries to:

1. Display all active products.
2. Display product name, brand, category, and selling price.
3. Display all customers from a selected city.
4. Display all premium customers.
5. Display customers registered within a date range.
6. Display products priced between two values.
7. Display products belonging to selected categories.
8. Display products whose names contain a specified keyword.
9. Display unique customer cities.
10. Display unique payment methods.
11. Count the total number of customers.
12. Count active and inactive products.
13. Count completed and cancelled orders.
14. Display the most expensive products.
15. Display the five products with the lowest stock.
16. Display employees ordered by salary.
17. Display recent orders using a result limit.
18. Display products with stock below reorder level.

19. Display failed payment attempts.
 20. Display overdue shipments.
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12. Aggregate Query Requirements

Prepare queries to:

1. Count customers by city.
 2. Count products by category.
 3. Calculate average product price by category.
 4. Calculate minimum and maximum product prices.
 5. Calculate total inventory value by warehouse.
 6. Count employees by department.
 7. Calculate average salary by department.
 8. Count orders by status.
 9. Calculate monthly order totals.
 10. Calculate monthly sales revenue.
 11. Calculate sales by warehouse.
 12. Calculate sales by product category.
 13. Calculate sales by brand.
 14. Calculate total discounts given.
 15. Calculate tax collected.
 16. Calculate total refunds.
 17. Calculate average customer order value.
 18. Calculate average supplier delivery delay.
 19. Count returns by return reason.
 20. Count shipments by delivery partner.
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13. GROUP BY and HAVING Requirements

Prepare reports to:

1. Display cities having more than five customers.
2. Display categories having more than ten products.
3. Display warehouses having inventory value above a specified amount.
4. Display customers who placed more than three orders.
5. Display products sold more than a specified quantity.
6. Display suppliers associated with more than five products.
7. Display delivery partners handling more than ten shipments.
8. Display employees processing more than five orders.
9. Display brands generating sales above a specified value.
10. Display months having sales above the annual monthly average.
11. Display products with an average rating above four.

12. Display warehouses having more than five low-stock products.
 13. Display suppliers with average delivery delay above seven days.
 14. Display customers receiving refunds above a specified amount.
 15. Display categories with return rates above a defined threshold.
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14. Join Requirements

Prepare queries demonstrating:

14.1 Inner Join

1. Products with categories
2. Products with brands
3. Orders with customers
4. Order items with product details
5. Payments with orders
6. Shipments with delivery partners
7. Returns with customers
8. Purchase orders with suppliers
9. Employees with warehouses
10. Inventory with products and warehouses

14.2 Left Join

1. All customers, including those without orders
2. All products, including those never ordered
3. All suppliers, including those without purchase orders
4. All warehouses, including those without stock transfers
5. All employees, including those who processed no orders
6. All orders, including orders without successful payments
7. All products, including products without reviews

14.3 Right Join

1. All warehouses, including those without employees
2. All categories, including those without products
3. All delivery partners, including those without shipments

14.4 Full Outer Join

1. Compare all ordered products and stocked products.
2. Compare supplier products and warehouse inventory.
3. Compare expected purchase order items and delivered supplier items.

14.5 Self Join

1. Display employees with their managers.
 2. Display managers and their direct reports.
 3. Display categories with their parent categories.
 4. Display employees earning more than their managers.
 5. Count employees reporting to each manager.
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15. Subquery Requirements

Prepare queries to:

1. Find products priced above the average product price.
 2. Find employees earning above the company average salary.
 3. Find employees earning above their department average.
 4. Find customers whose spending is above average.
 5. Find the most expensive product.
 6. Find the second-most expensive product.
 7. Find the customer with the highest total spending.
 8. Find the warehouse with the highest inventory value.
 9. Find products that have never been ordered.
 10. Find customers who have never placed an order.
 11. Find orders without successful payment.
 12. Find suppliers whose average price is below the overall supplier average.
 13. Find products with stock below the average warehouse stock.
 14. Find employees who processed more orders than the average employee.
 15. Find categories generating above-average revenue.
 16. Find delivery partners with delayed deliveries above average.
 17. Find customers who returned more products than the average customer.
 18. Find the supplier with the highest fulfilled purchase order value.
 19. Find warehouses with no pending stock transfers.
 20. Find products supplied by more than one supplier.
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16. UNION Requirements

Prepare queries to:

1. Combine customer names and employee names.
2. Combine supplier cities and customer cities.
3. Combine incoming supplier deliveries and outgoing shipments.
4. Combine product stock increases and stock decreases.
5. Combine successful payments and successful refunds.
6. Demonstrate the difference between UNION and UNION ALL.

7. Produce a common activity report containing:

8. Order created

9. Payment completed

10. Shipment dispatched

11. Return requested

12. Refund completed

17. Mathematical Function Requirements

Use mathematical functions to:

1. Calculate product price after tax.
2. Calculate discounted selling price.
3. Calculate gross order item amount.
4. Calculate net order item amount.
5. Calculate warehouse inventory value.
6. Round monetary values to two decimal places.
7. Calculate discount differences.
8. Calculate profit amount.
9. Calculate profit percentage.
10. Calculate refund percentage.
11. Calculate return rate.
12. Calculate warehouse capacity utilization.
13. Calculate supplier fulfilment percentage.
14. Calculate delivery success rate.
15. Calculate average days required to deliver orders.

18. String Function Requirements

Use string functions to:

1. Convert customer names to uppercase.
2. Convert email addresses to lowercase.
3. Display product name length.
4. Combine customer name, city, and phone.
5. Extract the first characters of a product code.
6. Replace selected words in product descriptions.
7. Remove unnecessary spaces.
8. Format supplier names using title case.
9. Extract the email domain.
10. Find the position of a character inside a string.

11. Generate display labels for products.
 12. Mask customer phone numbers.
 13. Mask parts of email addresses.
 14. Combine warehouse code and product code.
 15. Generate human-readable order references.
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19. Date and Timestamp Requirements

Use PostgreSQL date and timestamp functionality to:

1. Display the current date.
 2. Display the current timestamp.
 3. Extract year, month, and day from an order timestamp.
 4. Calculate customer account age.
 5. Calculate employee experience.
 6. Calculate supplier delivery delay.
 7. Calculate delivery duration.
 8. Calculate return request delay after delivery.
 9. Calculate refund processing duration.
 10. Determine coupon validity.
 11. Generate expected delivery dates.
 12. Group orders by day, week, month, quarter, and year.
 13. Format timestamps for reports.
 14. Identify orders placed during weekends.
 15. Identify orders placed during business hours.
 16. Find products not sold in the last six months.
 17. Find customers inactive for more than one year.
 18. Find overdue purchase orders.
 19. Find shipments delayed beyond expected delivery.
 20. Find refunds pending beyond the accepted processing period.
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20. Business Transaction Requirements

20.1 Customer Order Processing

When a customer places an order:

1. Validate that the customer account is active.
2. Validate the shipping address.
3. Validate that all products are active.
4. Validate available warehouse stock.
5. Reserve product quantities.
6. Record order item prices.

7. Apply eligible discounts.
8. Calculate tax.
9. Apply coupon conditions.
10. Calculate final order amount.
11. Create the order.
12. Create payment attempt records.
13. Update order status after successful payment.

20.2 Shipment Processing

When an order is shipped:

1. Identify the warehouse fulfilling each item.
2. Create one or more shipments.
3. Assign a delivery partner.
4. Generate a tracking number.
5. Add shipment items.
6. Reduce reserved stock.
7. Reduce available warehouse stock.
8. Update order item status.
9. Update order status to partially shipped or shipped.

20.3 Return Processing

When a customer requests a return:

1. Validate that the order item was delivered.
2. Validate the return eligibility period.
3. Validate the requested return quantity.
4. Record the return reason.
5. Approve or reject the request.
6. Receive the item at a warehouse.
7. Inspect the returned item.
8. Decide whether it can be restocked.
9. Calculate the refund amount.
10. Create the refund record.
11. Update return status.
12. Update inventory when applicable.

20.4 Supplier Procurement

When stock reaches reorder level:

1. Identify eligible suppliers.
2. Compare purchase price.
3. Compare supplier rating.
4. Compare average delivery duration.

5. Generate a purchase order.
6. Obtain approval.
7. Track supplier delivery.
8. Record accepted and rejected quantities.
9. Update warehouse stock.
10. Update purchase order status.

20.5 Stock Transfer

When stock is transferred:

1. Select source and destination warehouses.
 2. Confirm that the warehouses are different.
 3. Validate source stock.
 4. Request transfer quantities.
 5. Approve the transfer.
 6. Dispatch products.
 7. Reduce source stock.
 8. Receive products at the destination.
 9. Increase destination stock.
 10. Record damaged quantities.
 11. Complete the transfer.
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21. Transaction Management Requirements

Participants should demonstrate transaction handling for:

1. Order creation and stock reservation
2. Payment success and order confirmation
3. Shipment creation and inventory reduction
4. Return approval and refund creation
5. Supplier delivery and stock addition
6. Warehouse stock transfer
7. Order cancellation and stock release

The design must ensure that partial operations are not permanently saved when an error occurs.

Examples:

- An order should not be created if stock reservation fails.
 - A payment should not mark an order as confirmed if payment recording fails.
 - A stock transfer should not reduce source stock unless the transfer record is created successfully.
 - A refund should not be completed if the refund amount is invalid.
 - Supplier stock should not be added if delivery validation fails.
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22. View Requirements

Create logical reporting views for:

1. Product catalogue details
2. Customer order summary
3. Order item sales details
4. Warehouse inventory status
5. Low-stock products
6. Successful payment summary
7. Shipment tracking summary
8. Customer return summary
9. Supplier purchase order summary
10. Monthly sales summary

Views should simplify complex joins for reporting users.

23. Index Requirements

Create suitable indexes for frequently searched columns such as:

- Customer email
- Customer phone
- Product code
- Product name
- Category ID
- Brand ID
- Supplier ID
- Order date
- Order status
- Customer ID in orders
- Product ID in order items
- Payment transaction reference
- Payment status
- Shipment tracking number
- Shipment status
- Return status
- Warehouse and product combination
- Purchase order status
- Employee manager ID

Participants must explain why each index is required.

24. Import Requirements

Participants must prepare CSV files and import:

1. Customers
2. Customer addresses
3. Products
4. Suppliers
5. Warehouse inventory
6. Employees
7. Delivery partners

Import validation must identify:

- Duplicate emails
- Duplicate product codes
- Invalid dates
- Missing mandatory fields
- Negative prices
- Invalid warehouse codes
- Invalid category references
- Incorrect phone numbers

Invalid records should be moved to staging or error tables.

25. Export Requirements

Export the following reports:

1. Active product catalogue
2. Low-stock product report
3. Customer master report
4. Monthly order summary
5. Monthly sales report
6. Failed payment report
7. Shipment delay report
8. Product return report
9. Supplier performance report
10. Warehouse inventory valuation

Exports should be generated in CSV format.

26. Backup and Restore Requirements

Participants must:

1. Create a full database backup.
 2. Create a schema-only backup.
 3. Create a data-only backup.
 4. Back up selected tables.
 5. Restore the database into a new test database.
 6. Verify table counts after restore.
 7. Verify primary and foreign key constraints.
 8. Verify indexes and views.
 9. Document backup and restore steps.
 10. Demonstrate Docker-based backup and restore when PostgreSQL is running in a container.
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27. Final Management Reports

Report 1: Executive Sales Dashboard

Required output:

- Total orders
- Completed orders
- Cancelled orders
- Gross sales
- Discounts
- Taxes
- Net sales
- Refunds
- Final realized revenue
- Average order value

Report 2: Monthly Sales Trend

Required output:

- Month
- Number of customers
- Number of orders
- Products sold
- Gross sales
- Discount amount
- Tax amount
- Refund amount

- Net revenue

Report 3: Warehouse Performance

Required output:

- Warehouse name
- Number of products stocked
- Available quantity
- Reserved quantity
- Damaged quantity
- Inventory value
- Orders fulfilled
- Shipments dispatched
- Average fulfilment time

Report 4: Product Performance

Required output:

- Product name
- Category
- Brand
- Quantity sold
- Sales revenue
- Discount given
- Return quantity
- Return rate
- Average rating
- Remaining stock

Report 5: Customer Value Report

Required output:

- Customer name
- Customer type
- Registration date
- Total orders
- Completed orders
- Total spending
- Average order value
- Total refunds
- Loyalty points
- Last order date

Report 6: Supplier Performance

Required output:

- Supplier name
- Products supplied
- Purchase orders
- Ordered quantity
- Accepted quantity
- Rejected quantity
- Average purchase value
- Average delivery delay
- Supplier fulfilment percentage
- Supplier rating

Report 7: Delivery Partner Performance

Required output:

- Delivery partner
- Total shipments
- Delivered shipments
- Failed deliveries
- Returned shipments
- Average delivery time
- Delayed shipments
- Delivery success rate

Report 8: Return and Refund Analysis

Required output:

- Return reason
- Number of returns
- Returned quantity
- Refund amount
- Average refund processing time
- Most returned products
- Categories with highest return rates

Report 9: Payment Analysis

Required output:

- Payment method
- Total attempts

- Successful payments
- Failed payments
- Success percentage
- Total successful amount
- Refunded amount
- Net realized amount

Report 10: Inventory Risk Report

Required output:

- Warehouse
- Product
- Available quantity
- Reserved quantity
- Reorder level
- Average monthly sales
- Estimated days of stock remaining
- Recommended reorder quantity
- Preferred supplier

28. Advanced Analytical Challenges

Participants must prepare solutions for the following:

1. Identify the top five customers by total spending.
2. Identify the top five products by profit.
3. Identify products having high sales but low stock.
4. Identify products having stock but no sales.
5. Identify customers with repeated payment failures.
6. Identify customers with unusually high return rates.
7. Identify products with high return rates and low ratings.
8. Identify suppliers with repeated delivery delays.
9. Identify warehouses with frequent stock shortages.
10. Identify employees processing the highest order value.
11. Identify delivery partners with the lowest success rate.
12. Identify orders split across multiple warehouses.
13. Identify orders delivered through multiple shipments.
14. Identify products received from suppliers but never sold.
15. Identify categories where discounts reduce profitability.
16. Identify customers eligible for premium membership.
17. Identify coupons used beyond their allowed limit.
18. Identify payment records with no related successful order.
19. Identify refund records exceeding valid refundable value.
20. Identify inconsistent warehouse stock records.

29. Data Quality Requirements

Participants must identify and correct:

1. Duplicate customers
2. Duplicate suppliers
3. Duplicate product codes
4. Invalid emails
5. Invalid phone numbers
6. Missing product categories
7. Missing warehouse mappings
8. Negative stock
9. Negative payment values
10. Order totals not matching item totals
11. Payments exceeding order totals
12. Refunds exceeding paid amounts
13. Shipped quantities exceeding ordered quantities
14. Returned quantities exceeding delivered quantities
15. Received supplier quantities exceeding purchase quantities
16. Transfers between identical warehouses
17. Orders without order items
18. Shipments without shipment items
19. Returns without return items
20. Active transactions referencing inactive master records

30. Case Study Exercise Phases

Phase 1: Requirement Analysis

Participants must:

- Identify entities
- Identify attributes
- Identify relationships
- Identify mandatory and optional fields
- Identify business constraints
- Identify transaction boundaries

Phase 2: Logical Database Design

Participants must:

- Prepare an ER diagram
- Define cardinalities
- Define primary keys
- Define foreign keys
- Resolve many-to-many relationships
- Define normalization levels

Phase 3: Physical Database Design

Participants must:

- Select PostgreSQL data types
- Define indexes
- Define constraints
- Define delete and update behavior
- Define timestamp and audit fields

Phase 4: Data Preparation

Participants must:

- Create realistic master data
- Create transaction data
- Include successful and failed scenarios
- Include partial deliveries
- Include partial shipments
- Include returns and refunds

Phase 5: Query Implementation

Participants must implement:

- Basic queries
- Filtering
- Sorting
- Aggregation
- Grouping
- Joining
- Subqueries
- UNION
- Self joins
- Date functions

- String functions
- Mathematical functions

Phase 6: Reporting

Participants must generate all required operational and management reports.

Phase 7: Data Import and Export

Participants must import master data and export management reports.

Phase 8: Backup and Restore

Participants must create and restore PostgreSQL backups.

31. Expected Participant Deliverables

Each participant or team should submit:

1. Requirement analysis document
 2. Entity list
 3. Attribute list
 4. ER diagram
 5. Normalized database design
 6. Table relationship document
 7. Constraint document
 8. DDL implementation file
 9. DML implementation file
 10. Sample data file
 11. Query exercise file
 12. Reporting query file
 13. CSV import files
 14. CSV export files
 15. Backup file
 16. Restore verification report
 17. Screenshots of major reports
 18. Data quality validation report
 19. Index justification report
 20. Final project presentation
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32. Evaluation Criteria

Evaluation Area	Weightage
Requirement understanding	5%
Entity and relationship design	10%
Database normalization	10%
Primary and foreign keys	10%
Constraints and validations	10%
Master and transaction data	5%
SELECT and filtering queries	5%
GROUP BY and HAVING	5%
Join operations	10%
Subqueries and UNION	5%
Functions and timestamps	5%
Transaction management	5%
Management reports	5%
Import and export	3%
Backup and restore	2%
Data quality analysis	3%
Documentation and presentation	2%

Total: 100%

33. Final Challenge

Design a complete report that presents the full lifecycle of a selected customer order.

The report must contain:

- Customer details
- Billing and shipping addresses
- Order details
- Ordered products

- Product category and brand
- Warehouse allocation
- Item prices
- Discounts
- Tax amounts
- Coupon usage
- Payment attempts
- Successful payment
- Shipment details
- Delivery partner
- Tracking number
- Delivery duration
- Return details
- Returned products
- Inspection result
- Refund details
- Final realized revenue
- Current order status

The report should work correctly for:

- Fully delivered orders
- Partially shipped orders
- Cancelled orders
- Orders with failed payments
- Orders with multiple payment attempts
- Orders with product returns
- Orders with partial refunds